

ML LAB EXPT – 10

Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Load the Wisconsin Breast Cancer dataset
data = pd.read_csv('breast_cancer.csv')

# Preprocessing:
# 1. Drop unnecessary columns (e.g., 'id', 'Unnamed: 32')
data = data.drop(['id', 'Unnamed: 32'], axis=1, errors='ignore')

# 2. Convert diagnosis to numerical labels (Malignant=1, Benign=0)
data['diagnosis'] = data['diagnosis'].map({'M': 1, 'B': 0})

# 3. Separate features and target variable
X = data.drop('diagnosis', axis=1) # Features
y = data['diagnosis'] # Target

# Standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Determine optimal k using the Elbow method
# iterate through different values of k (from 1 to 10), applies k-means
# clustering and calculate the Within-Cluster Sum of Squares (WCSS).
# WCSS is a measure of the variance within each cluster.

wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300,
                    n_init=10, random_state=0)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
plt.plot(range(1, 11), wcss)
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

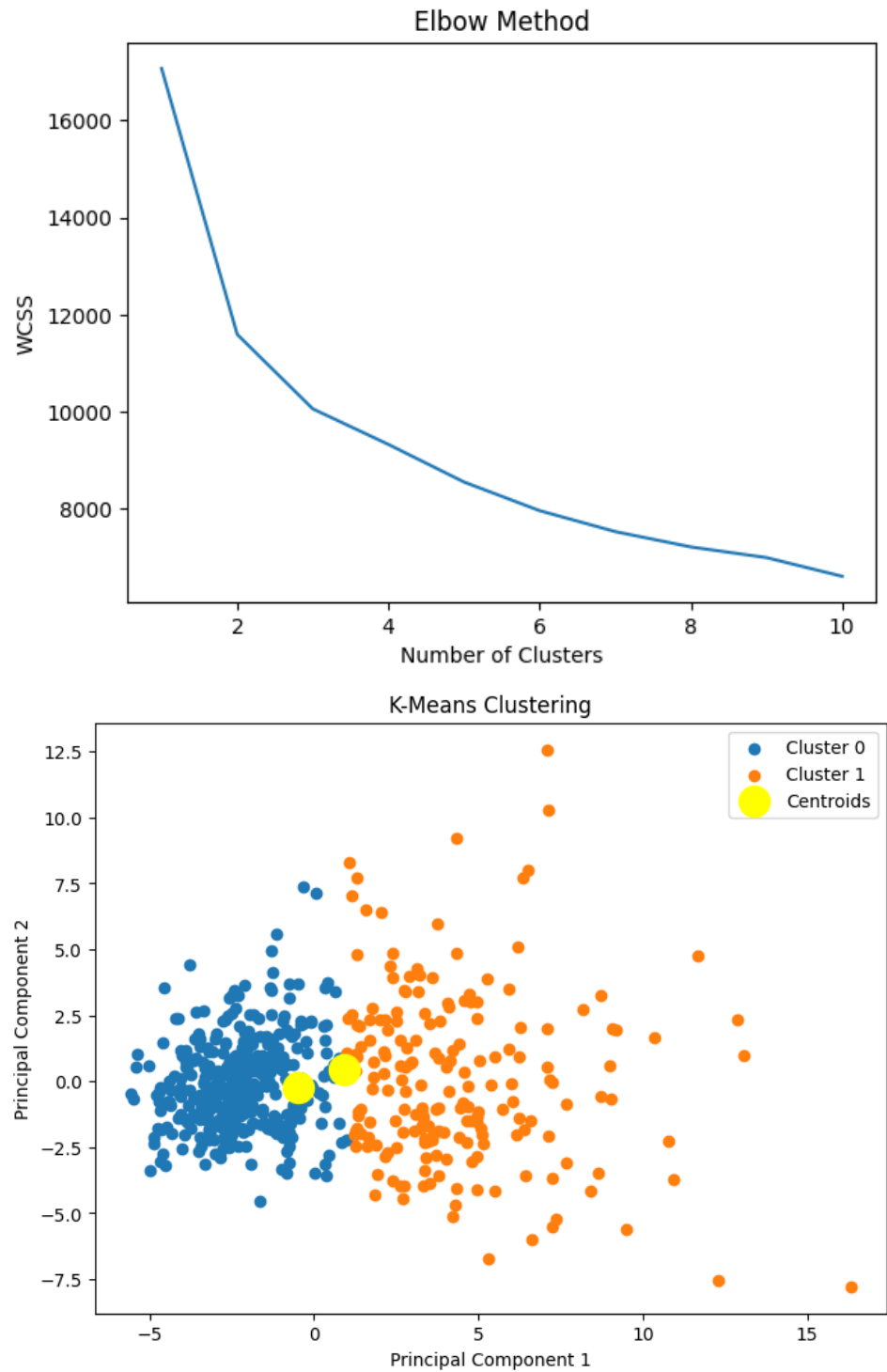
```
# Apply K-Means clustering (choose k based on the elbow method)
k = 2 #2 clusters
kmeans = KMeans(n_clusters=k, init='k-means++', max_iter=300, n_init=10,
random_state=0)
y_kmeans = kmeans.fit_predict(X_scaled)

# Visualize the clusters (using PCA for dimensionality reduction)
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(8, 6))
for i in range(k):
    plt.scatter(X_pca[y_kmeans == i, 0], X_pca[y_kmeans == i, 1],
                label=f'Cluster {i}')
plt.scatter(kmeans.cluster_centers_[0], kmeans.cluster_centers_[1],
            s=300, c='yellow', label='Centroids')

plt.title('K-Means Clustering')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend()
plt.show()
```

Output:



Model Evaluation:
Accuracy: 0.96

Classification Report:

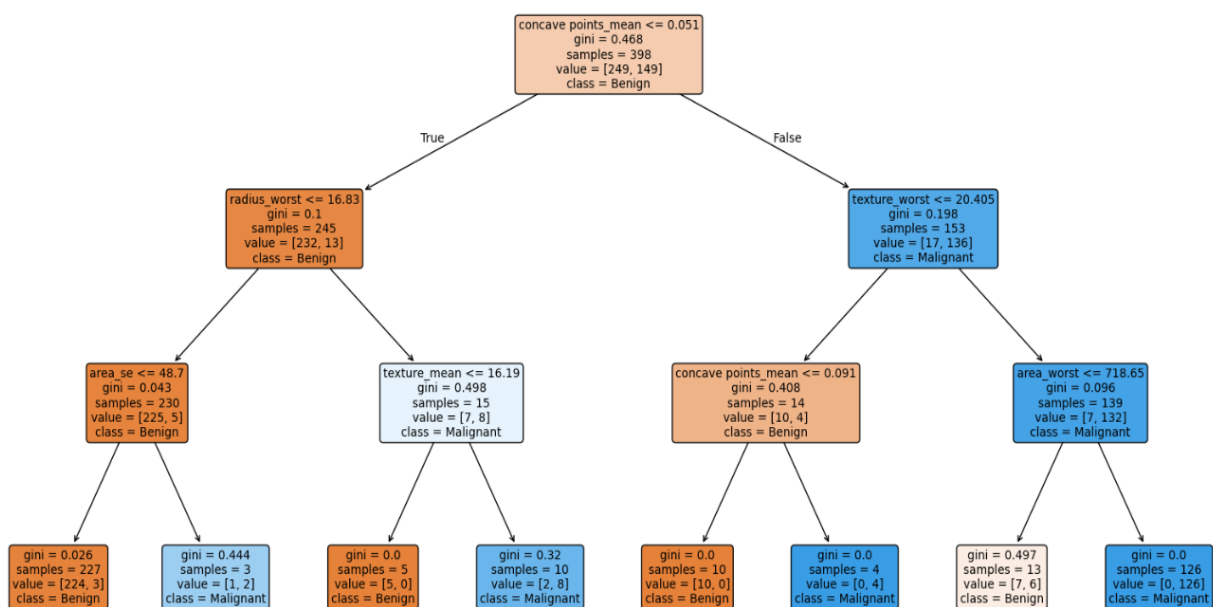
	precision	recall	f1-score	support
0	0.96	0.98	0.97	108
1	0.97	0.94	0.95	63
accuracy			0.96	171
macro avg	0.97	0.96	0.96	171
weighted avg	0.96	0.96	0.96	171

Confusion Matrix:
[[106 2]
[4 59]]

Decision Tree Rules:

```
--- concave points_mean <= 0.05
|--- radius_worst <= 16.83
|   |--- area_se <= 48.70
|   |   |--- class: 0
|   |   |--- area_se > 48.70
|   |       |--- class: 1
|   |--- radius_worst > 16.83
|   |--- texture_mean <= 16.19
|   |   |--- class: 0
|   |   |--- texture_mean > 16.19
|   |       |--- class: 1
|--- concave points_mean > 0.05
|   |--- texture_worst <= 20.40
|   |   |--- concave points_mean <= 0.09
|   |   |   |--- class: 0
|   |   |   |--- concave points_mean > 0.09
|   |   |       |--- class: 1
|   |   |--- texture_worst > 20.40
|   |   |--- area_worst <= 718.65
|   |   |   |--- class: 0
|   |   |   |--- area_worst > 718.65
|   |   |       |--- class: 1
|   |--- area_worst > 718.65
|   |   |--- class: 1
```

Decision Tree for Breast Cancer Classification



New Sample Classification:

Predicted class: Malignant

Probability: Benign=0.00, Malignant=1.00